Learning Objective 9.1: I understand the process of decoupling and finding solutions for Maxwell's equations for dynamic fields.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Vector phasors.** In a lossless, source-free region, the electric field is

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\, 5\cos(\omega t - \beta z + 30^\circ) + \hat{\mathbf{y}}\, 3\sin(\omega t - \beta z) \text{ V/m}$$

(a) Write the vector phasor $\widetilde{\mathbf{E}}(z)$.

Write the sine as a cosine, $\sin\theta = \cos(\theta - 90^\circ)$, so each component is a cosine with its own amplitude and phase:

$$\begin{aligned}\widetilde{\mathbf{E}}(z) &= \hat{\mathbf{x}}\, 5e^{j30^\circ}e^{-j\beta z} + \hat{\mathbf{y}}\, 3e^{-j90^\circ}e^{-j\beta z} \\ &= \left(\hat{\mathbf{x}}\, 5e^{j30^\circ} - j\,\hat{\mathbf{y}}\, 3\right)e^{-j\beta z} \text{ V/m}\end{aligned}$$

(b) Starting from $\nabla\times\mathbf{H} = \mathbf{J} + \partial\mathbf{D}/\partial t$ in the time domain, derive the phasor form of the Maxwell-Ampere law in this region. State each assumption where you use it.

Substitute $\mathbf{H} = \text{Re}\left[\widetilde{\mathbf{H}}e^{j\omega t}\right]$ and $\mathbf{D} = \text{Re}\left[\widetilde{\mathbf{D}}e^{j\omega t}\right]$. The curl acts on position, so it passes through the real part and $e^{j\omega t}$; the time derivative acts only on $e^{j\omega t}$:

$$\begin{aligned}\text{Re}\left[\left(\nabla\times\widetilde{\mathbf{H}}\right)e^{j\omega t}\right] &= \text{Re}\left[\left(\widetilde{\mathbf{J}} + j\omega\widetilde{\mathbf{D}}\right)e^{j\omega t}\right] \\ \nabla\times\widetilde{\mathbf{H}} &= \widetilde{\mathbf{J}} + j\omega\widetilde{\mathbf{D}}\end{aligned}$$

The equation holds at every instant, so the complex vectors are equal. Lossless, $\sigma = 0$, so $\widetilde{\mathbf{J}} = \sigma\widetilde{\mathbf{E}} = 0$ (no impressed current either, since the region is source-free). A simple medium has $\widetilde{\mathbf{D}} = \epsilon\widetilde{\mathbf{E}}$:

$$\nabla\times\widetilde{\mathbf{H}} = j\omega\epsilon\widetilde{\mathbf{E}}$$

2. **Decoupling the magnetic field.**

(a) Starting from the Maxwell-Ampere law, in phasor form, in a lossless, source-free region, derive the Helmholtz equation for the magnetic field. Name each of Maxwell's equations where you use it.

Take the curl of the Maxwell-Ampere law. The constant $j\omega\epsilon$ comes outside the curl:

$$\nabla \times \left(\nabla \times \widetilde{\mathbf{H}}\right) = j\omega\epsilon \, \nabla \times \widetilde{\mathbf{E}}$$

Faraday's law gives $\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}$. The identity $\nabla \times \nabla \times \mathbf{F} = \nabla\left(\nabla \cdot \mathbf{F}\right) - \nabla^2\mathbf{F}$ expands the left side, and Gauss's law for magnetism gives $\nabla \cdot \widetilde{\mathbf{H}} = 0$:

$$\nabla\left(\nabla \cdot \widetilde{\mathbf{H}}\right) - \nabla^2\widetilde{\mathbf{H}} = j\omega\epsilon\left(-j\omega\mu\widetilde{\mathbf{H}}\right) = \omega^2\mu\epsilon\,\widetilde{\mathbf{H}}$$

$$\nabla^2\widetilde{\mathbf{H}} + \beta^2\widetilde{\mathbf{H}} = 0, \qquad \beta = \omega\sqrt{\mu\epsilon}$$

Gauss's law for the electric field is the one equation this derivation does not use; the derivation of the electric-field equation uses it instead.

(b) On a transmission line, which line quantities play the parts of μ and ϵ, and which plays the part of $\widetilde{\mathbf{H}}$?

The inductance per meter L' plays μ, the capacitance per meter C' plays ϵ, and the current $\widetilde{I}$ plays $\widetilde{\mathbf{H}}$, as the voltage $\widetilde{V}$ plays $\widetilde{\mathbf{E}}$.

3. **The Helmholtz equation.** A lossless dielectric has $\epsilon_r = 4$ and $\mu_r = 1$. The frequency is 300 MHz.

   (a) Find β, the wavelength, and the phase velocity.

$$\beta = \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{2\pi \times 3 \times 10^8 \times 2}{2.998 \times 10^8} = 12.57 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = 0.500 \text{ m}, \qquad v_p = \frac{c}{\sqrt{\epsilon_r}} = 1.50 \times 10^8 \text{ m/s}$$

β = 12.57 rad/m

λ = 0.500 m

v_p = 1.50×10^8 m/s

   (b) Show that $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\, E_0 e^{-j\beta z}$ satisfies both the Helmholtz equation and $\nabla \cdot \widetilde{\mathbf{E}} = 0$.

Only the x component is nonzero, and it depends only on z:

$$\nabla^2 \widetilde{\mathbf{E}} = \hat{\mathbf{x}} \frac{d^2}{dz^2}\left(E_0 e^{-j\beta z}\right) = \hat{\mathbf{x}}\,(-j\beta)^2 E_0 e^{-j\beta z} = -\beta^2 \widetilde{\mathbf{E}}$$

$$\nabla \cdot \widetilde{\mathbf{E}} = \frac{\partial}{\partial x}\left(E_0 e^{-j\beta z}\right) = 0$$

   (c) The field $\widetilde{\mathbf{E}} = \hat{\mathbf{z}}\, E_0 e^{-j\beta z}$ also satisfies the Helmholtz equation. Explain why it cannot exist in a source-free region.

Its divergence is $\partial \widetilde{E}_z / \partial z = -j\beta E_0 e^{-j\beta z} \neq 0$, but Gauss's law with $\widetilde{\rho}_v = 0$ requires $\nabla \cdot \widetilde{\mathbf{E}} = 0$. A field that varies only along z cannot point along z in a source-free region.

4. **The speed of the wave and the line.**

(a) Use $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m to find the speed of a wave in free space.

$$v_p = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12}}}$$
$$= 2.998 \times 10^8 \text{ m/s}$$

v_p = 2.998×10^8 m/s

(b) A coaxial line has $L' = 250$ nH/m and $C' = 100$ pF/m, and its dielectric has $\mu_r = 1$. Using $L'C' = \mu\epsilon$ for the coax (Lesson 30), find ϵ_r and the phase velocity.

$$\epsilon_r = \frac{L'C'}{\mu_0 \epsilon_0} = \frac{250 \times 10^{-9} \times 100 \times 10^{-12}}{1.113 \times 10^{-17}} = 2.25$$

$$v_p = \frac{1}{\sqrt{L'C'}} = 2.00 \times 10^8 \text{ m/s}$$

The dielectric is polyethylene.

ϵ_r = 2.25

v_p = 2.00×10^8 m/s

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